A-A-54273
31 October 1990

COMMERCIAL ITEM DESCRIPTION

CEMENT, ZINC OXIDE AND EUGENOL, DENTAL
(Intermediate Restorative Material)

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for Military Specification MIL-C-36757A which is cancelled.

This commercial item description covers zinc oxide and eugenol dental cement suitable for use as an intermediate restorative material.

Salient characteristics:

The cement package shall consist of a minimum of 38.0 grams (g) of ivory colored powder, a minimum of 14.0 milliliters (ml) of liquid, a plastic dropper, a plastic powder scoop, and a set of instructions for use.

Powder. The composition of the cement powder shall be essentially zinc oxide, a zinc acetate accelerator, and high molecular weight polymethyl methacrylate. Cement powder shall contain no abrasive fillers such as quartz or alumina. The quality of the materials shall be as follows:

a. Zinc oxide shall be in accordance with the U.S.P. for lead and arsenic content only.

b. Zinc acetate shall be U.S.P.

c. The polymethyl methacrylate shall be nontoxic. The manufacturer shall submit a statement to the contracting officer that the polymer furnished has no toxic effect on the normal, healthy person.

Liquid. The composition of the liquid shall be eugenol, U.S.P. with acetic acid, A.C.S. added as an accelerator.

Accessories. The powder scoop shall be fabricated of clear styrene and shall deliver 0.20 to 0.24 grams of powder.

AMSC N/A FSC 6520

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:32Z
Check the source to verify that this is the current version before use.

Instructions. Adequate and accurate instructions for using the dental cement shall accompany each unit package. In addition, the instructions shall include setting time, mixing time, powder-liquid ratio, and methods for proportioning and manipulating the cement, including number of drops of liquid required to mix with one scoop of powder to make a mix of 5.2 to 6.3 powder-liquid ratio by weight. Two copies of the instructions shall be forwarded to the contracting officer for information purposes.

The dental cement shall be suitable as a sedative intermediate restorative material which may also be used as a base under amalgam restorations. The cement shall not form lumps, granules, or evolve gas when mixed according to the manufacturer's instructions.

Physical properties. The cement shall have the following physical properties when tested as specified below. The setting time shall be 7.5 to 9.5 minutes. The compressive strength shall be 9000 pounds per square inch, minimum. The solubility and disintegration shall be 0.12 percent by weight, maximum. The film thickness shall be 50 to 90 microns (0.0020 to 0.0035 inches). The consistency shall be 16.5 to 19.5 millimeters.

Immediate container. Immediate container for the cement powder shall be a plastic bottle. Immediate container for the liquid shall be an amber glass bottle with screw cap closure. Each bottle shall be permanently and legibly marked with the lot number, date of manufacture, and inspection test date. Each bottle shall be further marked with the following legend:

"STORE AT CONTROLLED ROOM TEMPERATURE 15°-30°C (59°-86°F)".

Shelf life. The shelf life of the cement shall be not less than 24 months from date of manufacture when stored at controlled room temperature (59° to 86°F).

Delivery schedule. Not more than 4 months shall have elapsed from the date of manufacture of the cement to the date of delivery to the Government.

Workmanship. The powder and liquid shall be free from all dirt, foreign materials, and impurities which detract from their appearance or which may impair their serviceability.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:32Z
Check the source to verify that this is the current version before use.

Unit. Package (PG). One package containing one bottle of powder, one bottle of liquid, a dropper, a powder scoop, and instructions for use, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for inspection. Sampling for inspection shall be conducted in accordance with MIL-STD-105. Where feasible, the same sample shall be used for the determination of two or more inspection characteristics. The unit of product for inspection purposes shall be one package of cement. Unless otherwise specified, sampling for examination shall be conducted at inspection level II, AQL 2.5, and sampling for tests shall be conducted at inspection level S-1, AQL 1.0.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:32Z
Check the source to verify that this is the current version before use.

Examination. The dental cement shall be examined for defects including, but not limited to, those listed below:

Defects.

Package does not contain all specified components.
Powder not homogeneous or free of extraneous material.
Liquid not clear or has significant color.
Bottle for liquid not amber or does not have screw cap closure.
Loose caps, broken bottles, leaks, etc. present.
Marking of immediate containers not as specified.
Instructions missing, incorrect, incomplete, or illegible.

Tests. Tests shall be conducted to determine compliance with specification requirements.

Preparation of test specimens. The preparation of test specimens shall be conducted at a temperature of 70 to 75°F, and 45 to 55 percent relative humidity with a powder-liquid ratio of 5.2 to 6.3 by weight (standard consistency).

Consistency. The procedure for the consistency test shall be as described in American Dental Association (ADA) Specification No. 30 for Dental Zinc Oxide-Eugenol Type Restorative Materials, except that a 520 gram load (500 gram weight and 20 gram glass plate) shall be applied 2 minutes after the mix is started. The load shall be applied for 8 minutes. Then consistency shall be measured.

Setting time. A mix of cement using 0.25 ml of liquid and an amount of powder in proportion shall be loaded into a polyethylene crown form cap with approximate internal dimensions of 9.5 mm diameter by 7 mm deep. Type A durometer hardness shall be determined as described in ASTM designation D2240. Readings shall be taken at 0.5 minute intervals. Hardness shall be plotted against time, and the setting time in minutes shall be the interpolated value of time when durometer reading equals 50. Setting time shall be reported to the nearest 0.5 minute.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:32Z
Check the source to verify that this is the current version before use.

Compressive strength. Procedure for compressive strength test shall be as described in ADA Specification No. 30. Test equipment used for determining compressive strength shall be the Instron Corporation's Test Machine, or other suitable equipment, provided prior approval is received from the contracting officer. Testing conditions on the Instron test machine shall be as follows:

a. Crosshead speed of 0.2 inch per minute.
b. Force range of 1000 pounds.
c. Chart speed of 10 inches per minute.

The value for compressive strength shall be reported as the average of three or more from a group of five specimens and shall be rounded off to the nearest 50 pounds. If the values for individual specimens fall more than 15 percent below the average of the five, they shall be discarded and the average of the remaining specimens shall be reported. In case more than two of the specimens are eliminated, the test shall be repeated.

Solubility and disintegration. Procedure for solubility and disintegration test shall be as described in ADA Specification No. 30.

Film thickness. Procedure for film thickness test shall be as described in ADA Specification No. 30.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producer's own drawings, specifications standards and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:32Z
Check the source to verify that this is the current version before use.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and of regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:32Z
Check the source to verify that this is the current version before use.

NOTE: The following National Stock Number is covered by this document:

National Stock Number	Item Identification
6520-00-181-7732	CEMENT, ZINC OXIDE AND EUGENOL, DENTAL, Intermediate Restorative, Ivory, 38 Gram

Ordering data. (Intermediate/exterior package quantities, labeling, and marking must be as specified in the contract and/or order).

MILITARY INTERESTS:

Custodians:
Army - MD
Navy - MS
AIR FORCE - 03

PREPARING ACTIVITY:

DoD-MB

AGENT:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:
VA-OSS
PHS
FDA-MPQAS

PROJECT NO. 6520-2362
LOCATION: WORD/CEMENTPAUL

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:32Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEFENSE LOGISTICS AGENCY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 4266 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEFENSE LOGISTICS AGENCY

Commander
Defense Personnel Support Center
ATTN: DPSC- RSTE
2800 South 20th Street
Philadelphia, Pennsylvania 19101-8419

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:32Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-54273	CEMENT, ZINC OXIDE AND EUGENOL, DENTAL

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 1426 PREVIOUS EDITION IS OBSOLETE

Source: https://assist.dla.mil – Downloaded: 2026-10-26T09:02Z
Check the source to verify that this is the current version before use.